

Project Name – End-to-End Retail Performance and Behavioral Analytics

Data Cleaning Steps :-

1.Data Loading and Initial Inspection

The dataset was imported into the Jupyter Notebook using Python libraries. Initial inspection was performed using functions such as `head()`, `tail()`, `info()`, and `describe()` to understand the structure, data types, and summary of the dataset.

2.identifying & removing duplicate values

Duplicate values were identified using `.duplicated().sum()` And Removing Duplicates using `drop_duplicates()` Detect and remove duplicate records to ensure data consistency.

3.Identifying Missing Values

Missing values were identified using built-in functions(`.isnull()`) to check null counts across all columns. This step helped in understanding which attributes required cleaning or further attention.

4. Data Type Review

The dataset's column data types were reviewed by using `.dtype()` to ensure numerical and categorical attributes were correctly interpreted. This step ensured the dataset was suitable for further analysis.

5. Handling Inconsistent and Null Values

Based on the initial analysis, missing and inconsistent values were reviewed and handled appropriately to maintain data integrity and avoid errors during analysis.ex:order_date,signup_date

6. Prepared Dataset for Analysis

After completing the above steps, the dataset was considered ready for analytical process and visualization tasks.

7. Final Data Validation

After cleaning, the dataset was revalidated to confirm:

- No missing values remained
- Correct data types across all columns
- no duplicate values remained

8. Basic Data Exploration

Exploratory Data Analysis (EDA) was carried out using visualizations such as distribution plots, count plots, and box plots.

9. Exporting Cleaned Data

The final cleaned dataset was saved and prepared for further analysis, visualization, and reporting in SQL, Power BI, and other analytics tools.